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ORIGINAL PAPER

Supply function equilibria for uniform price auction in oligopolistic markets

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Abstract We consider game-theoretic models related to the supply function auction for electricity markets. We determine the set of supply function equilibria (SFE), introduced by Klemperer and Mayer (Econometrica 57:1243–1277, 1989), for a symmetric oligopoly with linear demand, fixed marginal cost and capacity constraint. This set depends on the maximum random shock of the demand function. We also study the best response dynamics and show that in general it does not converge to any SFE. We find out sufficient conditions for the convergence and conclude on the optimal parameters of the auction.

Keywords Supply function equilibrium · Best response dynamics · Wholesale electricity market · Oligopoly · Auctions

Mathematics Subject Classification 91A40 · 91B26 · 91A80

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1 Introduction

Many wholesale electricity markets connect several neighbouring countries. Each country is interested in an efficient design and performance of the market in particular of limitation of market power of large producers. This can be considered as a platform for international collaboration. OR methods are widely applied in analysis of problems related to organization and development of the markets. Typically their basic components—day-ahead and real-time markets—perform as uniform price auctions where a producer's bid is a monotonous function that determines the supplied quantity depending on the price. A market price in such an auction equalizes an aggregate supply function of all firms and a demand function.

A rather high concentration of generation is typical for electricity markets. Consumers usually do not play an active role in electricity auctions. Their behavior corresponds to a known demand function with a low elasticity. So market designers face an important problem to restrain giant producers' market power. Standard measures of anti-trust policy such as splitting the market are often inapplicable, because they reduce reliability and economy of scale, which are important in electricity industry. Another way is to design such a market mechanism that its equilibrium state is sufficiently close to the Walrasian equilibrium—the optimal state of the market according to the Welfare theorem (Debreu 1954).

The real electricity auctions differ in the rules for acceptable bids. Russian day-ahead market (DAM) accepts piece-wise constant bids with at most 3 steps, a different bid for every hour of the next day, while the market of England and Welsh at some time permitted up to 48 steps, but a unique bid for the whole day. Some auctions accept not only piece-wise constant, but also piece-wise linear bids. Determination of the optimal auction rules is an interesting Operations Research problem. Theoretical studies show that the rules essentially influence the properties of the uniform price auction. Vasin et al. (2007) consider a model corresponding to Russian DAM. For each round of the auction, the demand function is fixed and a strategy of each commodity producer is a non-decreasing step function with a limited number of steps. The paper shows that, for any Nash equilibrium, the market price lies between the competitive equilibrium price and the Cournot price, and vice versa, each price in this range corresponds to a Nash equilibrium. However, only the Nash equilibrium corresponding to the Cournot outcome is stable with respect to adaptive dynamics for the repeated auction. Thus, the estimate of the market power for this variant is the same as for the Cournot competition.

Variations of demand within a period for which bids are submitted plays a crucial role in the analysis of the second variant of the auction. A corresponding model was introduced by Klemperer and Meyer (1989) and developed in many papers (see Anderson and Hu 2008; Anderson and Philpott 2002; Baldick et al. 2000; Green 1999; Green and Newbery 1992; Holmberg et al. 2008; Newbery 1998, 2002, 2008; Wang et al. 2007). The original model permits a bid of a producer to be a smooth function of a price, whilst a demand is a function of an uncertain parameter besides the price. For each parameter value, the market clearing price is determined from the balance of the aggregate supply function and the current demand function. A bid profile is called *a supply function equilibrium (SFE)* if, for any demand realization, the bid of each firm maximizes its profit under fixed bids of other producers. For a symmetric

oligopoly, Klemperer and Meyer (1989) identify first order condition for a bid to be an equilibrium in a form of a differential equation and portray the set of SFE. Green and Newbery (1992) obtain an explicit formula of the SFE for a symmetric oligopoly with linear marginal cost and demand functions. Baldick et al. (2000) generalize this result for an asymmetric linear model. Anderson and Hu (2008) and Holmberg (2009) provide methods for calculation of the SFE for a certain class of asymmetric non-linear models. Abolmasov and Kolodin (2002) and Dyakova (2003) apply this approach to study electricity markets in two Russian regions. They use the affine approximations of the true supply functions and obtain a significant decrement of the "market power" in the supply function auction compared with the Cournot auction. Holmberg and Newbery (2010) mention several other markets with a similar effect.

Proceeding from these results, one may conclude that the English variant of the auction is always better than the Russian one. However, except for two studies of Texas real-time market (Hortascu and Puller 2008; Sioshansi and Oren 2007), there is no direct evidence of producers setting SFE bids. Theoretical results show that their computation is in general a non-trivial mathematical task. The solution depends upon common knowledge on the demand function and expenditures of all rivals. Moreover, submission of such a bid is rational only if behavior of other agents in the auction corresponds to the SFE. All the above mentioned complexity should be weighed against the possibility of real agents' performance to comply with this conception. Akin issues for Nash equilibria (NE) usage in economics are addressed by analysis of adaptive learning models. Milgrom and Roberts (1991) describe behavior dynamics and demonstrate that, under certain conditions, it converges to a stable Nash equilibrium under incomplete information of players on rivals' pay-off functions and bounded rationality.

The present paper considers a similar approach for the uniform price auction with uncertain demand. As many other studies, we consider the auction, assuming that consumers have no market power and bid as in a competitive market (Holmberg and Newbery 2010 justify this assumption). Following Rudkevich (1999) and Anderson and Hu (2008), we study existence and dynamics of *strong best responses*: a bid of a producer is a strong best response (SBR) to a bid profile of other participants if it maximizes his profit under any value of the uncertain parameter. We examine two variants of a symmetric oligopoly with a linear demand function: a) with linear marginal cost, b) with fixed marginal cost and capacity constraint. Our purpose is to find out for each case whether the dynamics exists and converges to any SFE. Vasin and Gusev (2011) described the SFE for a duopoly with linear and fixed marginal cost functions and studied existence of the best responses under fixed marginal costs and fixed random parameter value. We generalize the results to an oligopoly and give a detailed discussion of the BRD in the fixed marginal cost case. Section 2 contains a general formal model and recalls main results by Klemperer and Meyer (1989). Section 3 specifies a result by Rudkevich (1999) on the convergence of the strong best response dynamics (SBRD) to the unique SFE for a model of oligopoly with linear marginal costs. Section 4 determines the set of SFE depending on the maximum demand value, and provides a comprehensive study of the SBRD for the symmetric oligopoly with constrained capacity and constant marginal cost. We determine a partitioning of the demand shocks set in such intervals that the SBRD exists if every time the shock values belong to the same interval. We show that the SBRD converges to the SFE if

the demand shocks are in the highest or the lowest interval. But for each interval in the intermediate area, the SBRD is cyclic with its special cycle length. So producers do not learn to submit SFE bids from their experience in this case. Section 5 discusses different results on applicability of the SFE concept for modeling electricity markets and concludes on the optimal choice of the uniform price auction parameters depending on the market structure. The preliminary results of our study were reported at the International Annual Conference of the German Operations Research Society 2012 (Vasin and Dolmatova 2014). In the present paper we provide complete proofs of all the results and analysis for the oligopoly with linear demand and marginal cost functions.

2 Supply functions auction and equilibrium

Consider a market with a set $N = \{1, 2, \ldots, n\}$ of players (producers) with identical cost functions $C(q)$, that depend on output q and meet conditions: $C'(q) > 0$, $C''(q) \geq 0$ for any $q \geq 0$. Demand $D(p, t)$ is a function of price $p \geq 0$, and also depends on a random factor t. The probability density of the factor is strictly positive for the set $[\underline{t}, \overline{t}]$. For any p, t, the demand function satisfies: $D_p < 0$, $D_{pp} \leq 0$, $D_{pt} = 0$, $D_t > 0$. A strategy of player i is a non-decreasing supply function $S^i(p)$ that maps produced volume to the market price p. Making their choices $S^i(p)$, the players do not know the value of the random factor t. When t is known, a strategy profile $\overrightarrow{S} = (S^1(p), S^2(p), \ldots, S^n(p))$ induces the price $p(\overrightarrow{S}, t)$ that balances the aggregate supply and demand: $D(p(t)) = \sum_{i=1}^{n} S^i(p(t))$. If there are several clearing prices or such a price does not exist, then no production takes place and the firms' profits are zero. Each producer i intends to maximize his pay-off function:

$$\pi_i(\overrightarrow{S}, t) = p(\overrightarrow{S}, t) S^i(p(\overrightarrow{S}, t)) - C(S^i(p(\overrightarrow{S}, t)), \quad i \in N.$$

Strategy profile $\overrightarrow{S}^* = (S^{*i}, i \in N)$ is called an *SFE* if, for any $t \geq 0$ and $i \in N$, $S^{*i} \in \mathrm{Argmax}_{S_i}(\pi_i(S^i, S^{*-i}, t))$.

Proposition 1 (See Proposition 1.2 in Klemperer and Meyer 1989) *If* $\sup_t D(0, t) = \infty$, *then* $\overrightarrow{S}^* = (S^{*1}, S^{*2}, \ldots, S^{*n})$ *is SFE if and only if* $S^i(p) \equiv S(p)\ \forall i \in N$, $S(p)$ *monotonously increases in* p *and meets equation:*

$$S'(p) = \frac{1}{n-1}\left[\frac{S(p)}{p - C'(S(p))} + D_p(p)\right]. \tag{1}$$

Below we deal only with monotonously non-decreasing bid functions as feasible ones. This is consistent with auction rules in practise.

3 Oligopoly with linear marginal costs

Consider a symmetric oligopoly with n producers, where a cost function of each producer is $C(q) = (c_0 + 0,5c_1 q)q$, $c_0 > 0, c_1 > 0$, and the demand function is

$D(p,t) = \overline{D}(t) - dp$, where $d > 0$ and $\overline{D}(t)$ is a maximal demand value depending on a random parameter t. According to Proposition 1, an equilibrium supply function for this case should meet the differential equation:

$$S'(p) = \frac{1}{n-1}\left(\frac{S(p)}{p-(c_0+c_1*S(p))} - d\right).$$

There exists the infinite set of nonlinear solutions. However, the condition $\sup_t \overline{D}(t) = \infty$ results in a unique and linear SFE (Klemperer and Meyer 1989):

$$S^*(p) = (p-c_0)\frac{n-2-c_1d+\sqrt{(n-2)^2+2nc_1d+c_1^2d^2}}{2c_1(n-1)}. \tag{2}$$

Let $\mathbf{S}^{-i}$ denote a strategy profile of all players except for i, $S^{-i} := \sum_{j\neq i} S^j(p)$ be the aggregate bid of these players. Bid $S^i(p)$ is called a strong best response for any bid $S^{-i}(p)$, if for each $t \in [\underline{t}, \overline{t}]$ solution $p(t)$ to $S^i(p) + S^{-i}(p) = D(p,t)$ provides the maximum profit:

$$p(t) \longrightarrow \max_p\{(D(p,t) - S^{-i}(p))p - C(D(p(t) - S^{-i}(p)))\}.$$

Consider the strong best response dynamics (SBRD) for the repeated auction with identical players. At every time $\tau = 1, 2, \ldots$, each firm sets a bid $S(p,\tau)$ that is a strong best response to its competitors' bids $S(p,\tau-1)$ at the previous time. We do not distinguish players in the dynamics since we assume $S(p,0) = 0$, so the best responses of all the players are the same.

Rudkevich (1999) analyzed the general case of linear market, derived the form of the learning process and showed convergence of the dynamics to the unique linear SFE of the model. For a symmetric case we obtain the following implication.

Proposition 2 *Let $\{\overline{D}(t), t \in T\} = [dc_0, \infty)$. Then, for a symmetric n-firm oligopoly model, the bid $S^i(p) = (p-c_0)\frac{d+(n-1)k}{1+c_1(d+(n-1)k)}$ is the unique strong best response to the competitors' bids $S^j = k(p-c_0)$, $j \in N\backslash\{i\}$.*

The best response at time τ is $S(p,\tau) = k_\tau(p-c_0)$, where $k_\tau = \frac{d+(n-1)k_{\tau-1}}{1+c_1(d+(n-1)k_{\tau-1})}$. The unique fixed point $k^* = \frac{n-2-c_1d+\sqrt{(n-2)^2+2nc_1d+c_1^2d^2}}{2c_1(n-1)}$ for this dynamics corresponds to the SFE (2) of the static auction model.

The relation for k_τ immediately follows from Lemma (2) (Rudkevich 1999) best response equations. By setting k_τ^* in both parts of this relation, we obtain a quadratic equation with the unique positive solution given in Prop. 2.

Proposition 3 *The SBRD converges to the SFE of the static model with a geometric rate : $|\frac{k_\tau}{k^*} - 1| \leq |\frac{k_1}{k^*} - 1|(1+cd)^{1-\tau}$.*

Proof According to Prop. 2, $k^* = \frac{d+(n-1)k^*}{1+c_1(d+(n-1)k^*)}$ and

$$\frac{k_\tau}{k^*} - 1 = \frac{\frac{d+(n-1)k_{\tau-1}}{d+(n-1)k^*} + c_1(d+(n-1)k_{\tau-1})}{1+c_1(d+(n-1)k_{\tau-1})}$$

$$= \frac{|k_{\tau-1} - k^*| \cdot \frac{n-1}{d+(n-1)k^*}}{1 + c(d + (n-1)k_{\tau-1})} - 1 \le |\frac{k_1}{k^*} - 1|(1 + cd)^{\tau-1}$$
$$= \frac{\frac{k_{\tau-1}-k^*}{k^*} \cdot \frac{(n-1)k^*}{d+(n-1)k^*}}{1 + c_1(d + (n-1)k_{\tau-1})}.$$

We obtain:

$$|\frac{k_\tau}{k^*} - 1| \le |\frac{k_\tau - 1}{k^*} - 1| \cdot \frac{1}{1 + c_1 d}\left(\frac{d}{(n-1)k^*} + 1\right)^{-1}$$
$$\le |\frac{k_1}{k^*} - 1|(1 + c_1 d)^{1-\tau}\left(\frac{d}{(n-1)k^*} + 1\right)^{1-\tau}.$$

Thus the increase of number of market participants reduces the speed of convergence of the BRD in this case.

4 Oligopoly with fixed marginal costs and capacity constraints

Consider a market with n symmetric suppliers, each characterized by the capacity constraint Q and the linear cost function $C(q) = cq$, where $q \le Q$ and $c > 0$. The equilibrium supply function profile is constructed as a set of continuous monotone functions, each satisfying the equation:

$$S'(p) = \frac{1}{n-1}\left(\frac{S(p)}{p-c} - d\right).$$

Newbery (2008) provides a general solution: $S(p, A) = A(p-c)^{1/(n-1)} - \frac{d(p-c)}{n-2}$, $n > 2$ depending on the integration constant A. The function reaches its maximum $q(A)$ under $p = p(A) \stackrel{def}{=} \frac{A(n-2)}{d(n-1)}^{\frac{n-1}{n-2}}$. The inverse function for $q(A)$ is $A(q) = d \cdot \left(\frac{q}{d}\right)^{\frac{n-2}{n-1}} \cdot \frac{n-1}{n-2}$. In case the capacity constraint is loosen or not essential, the balance condition $nS(p, A) = \overline{D} - dc$ sets the equilibrium price $\tilde{p}(A, \overline{D})$. Denote $D^* = \sup_t\{\overline{D}(t) - dp\}$.

Theorem 1 *If the maximum demand value $D^* \ge (n+1)Q$, then there exists a unique SFE in the oligopoly model. The equilibrium bid is*

$$S^*(p) = \begin{cases} S(p, A(Q)) & \text{for } c \le p \le p(A(Q)), \\ Q & \text{for } p \ge p(A(Q)). \end{cases} \tag{3}$$

If $(n-1)Q < D^* < (n+1)Q$, then bid (see bid type 1 Fig. 1) $\overline{S}(p, A) = \begin{cases} S(p, A), & \text{if } p \le p(A), \\ S(p(A), A), & \text{if } p \ge p(A) \end{cases}$ determines SFE, for any $A \in (A(D^*/(n+1)), A(Q))$.

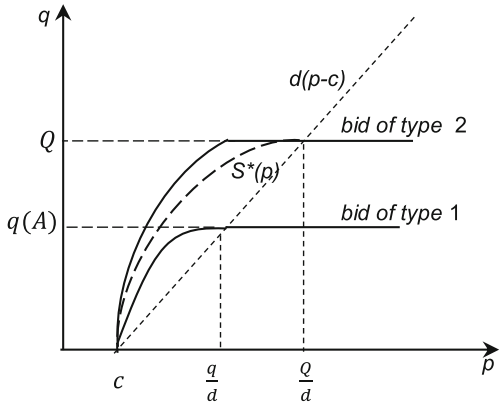


Fig. 1 Two types of possible equilibrium bids

Bid (see bid type 2 Fig. 1) $\overline{\overline{S}}(p, A) = \min\{\overline{S}(p, A), Q\}$ *also determines SFE, for any* $A \in (A(Q), \overline{A}(D^*))$, *where* $\overline{A}(D^*)$ *is the solution to*

$$S(\tilde{p}(\overline{A}, D^*))(\tilde{p}(\overline{A}, D^*) - c) = \frac{(D^* - (n-1)Q)^2}{4d} + c\frac{(D^* - (n-1)Q)}{2}. \quad (4)$$

If $D^* < (n-1)Q$, *then the capacity constraint is not binding for feasible bids. In this case,* $\forall A > A(D^*/(n+1))$, *the bid* $S(p, A)$ *determines SFE. For* $A \to \infty$, *the SFE outcome tends to the Walrasian equilibrium.*

Proof Consider two types of possible equilibrium bids illustrated in Fig. 1.

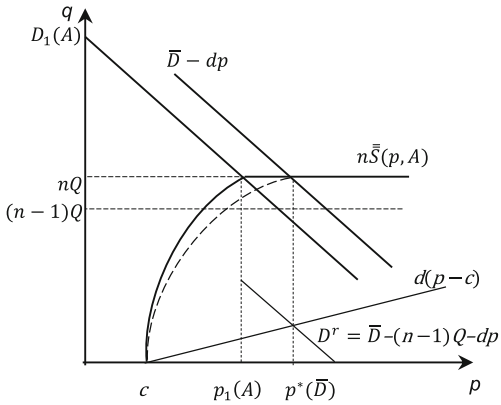
A bid of the first type corresponds to the value $A < A(Q)$. The curve $S(p, A)$ reaches its maximum value $q < Q$ for $p = q/d + c$, and then the bid stays constant. If $D^* > (n+1)q$, then for a maximum demand value the price exceeds $q/(d(n-1)) + c$, the competitors' bid is constant in the neighborhood of this price, and the best response to it is determined by the Cournot supply function $S_C(p) = \min\{(p-c)d, Q\} > q$. So that bid is not an equilibrium for this case. If $D^* \leq (n+1)q$ and the competitors bid according to type 1, then the residual demand function is

$$D^2(p) = \begin{cases} \max\{0, \overline{D} - dp - (n-1)S(p, A)\} & \text{for } p \leq q/d(n-1) + c, \\ \max\{0, \overline{D} - dp - (n-1)q\} & \text{for } p \geq q/d(n-1) + c. \end{cases}$$

Since the function $S_C(p)$ exceeds the residual demand in the last interval, then for $p \geq q/d(n-1) + c$ the profit reaches the maximum for $p = q/(d(n-1)) + c$. According to Proposition 1 in Klemperer and Meyer (1989), in the first interval the profit function reaches its maximum for $\overline{D} - dp = nS(p, A)$. Thus, for $D^* \leq (n+1)q$ the bid $\overline{S}(p, A(q))$ determines SFE.

The second type bid $\overline{\overline{S}}$ corresponds to values $A > A(Q)$ (Fig. 2). Let us determine the values $p_1(A)$, $D_1(A)$ from $D_1(A) - dp_1(A) = nQ = nS(p_1(A), A)$. Consider a residual demand function for $\overline{D} \in (D_1(A), (n+1)Q)$. The profit increases for values p in the interval $[0, p_1(A)]$. For $p \geq p_1(A)$ the profit function reaches its maximum

Fig. 2 Equilibrium of the type 2



in the point $p^*(\overline{D})$ of the residual demand function and the Cournot supply schedule intersection: $d(p^* - c) = \overline{D} - (n-1)Q - dp^* \Rightarrow p^*(\overline{D}) = (\overline{D} - (n-1)Q)/2d + c$. Thus $\overline{S}(p, A)$ is not SFE, if $D^* \geq D_1(A)$.

For $\overline{D} \in (0, D_1(A))$ the residual demand intersects $\bar{\bar{S}}(p, A)$. In the interval $[0, p_1(A)]$, the profit function reaches its maximum at the intersection price $\tilde{p}(A, \overline{D})$: $\overline{D} - dp = nS(p, A)$. For $p \geq p_1(A)$, it is at $p^*(\overline{D})$ if $p^*(\overline{D}) > p_1(A)$. However, for $\overline{D} \leq Q + ndp_1(A)$, the last maximum is at $p_1(A)$ and is clearly less than the first one.

Thus $\bar{\bar{S}}(p, A)$ is not SFE, if $\overline{D} > D_2(A) := Q + ndp_1(A)$.

Finally, we consider the interval $(D_2(A), D_1(A))$. For $p \geq p_1(A)$, the profit maximum is $M_1(\overline{D}) = \frac{(\overline{D}-(n-1)Q)^2}{4d}$, and for $p \leq p_1(A)$, $M_2(\overline{D}) = S(\tilde{p} \cdot (A(\overline{D})))(\tilde{p}(A(\overline{D})) - c)$. There exists a unique value $\overline{D}(A) \in (D_2(A), D_1(A))$, for which they are equal. Indeed, the derivative $M_2'(\overline{D}) - M_1'(\overline{D}) = \tilde{p}(A, \overline{D}) - c - (\overline{D} - (n-1)Q)/2d$ is negative for $\overline{D} = D_1(A)$ and increases in this interval, so it is negative in it. Thus, the bid $\bar{\bar{S}}(p, A)$ determines SFE, if $D^* \leq \overline{D}(A)$. (For $D^* < (n-1)Q$, the capacity constraint is not binding, and the statement follows directly from Klemperer and Meyer 1989). The inverse function $\overline{A}(\overline{D})$ meets Eq. (4) and is monotonous. This completes the proof.

The strong best response dynamics for the model with fixed marginal cost and no capacity constraint ($C(q) = cq, c > 0, q > 0$) is a particular case of the model in Sect. 3. Without the loss of generality, assume $c = 0$. Positive marginal cost will only cause the consecutive shift $(p - c)$ in equations. The best response dynamics for n firms can be described as $S(p, \tau) = k(\tau)p$, where the coefficient at time τ is $k(\tau) = d\sum_{s=0}^{\tau-1}(n-1)^s$. For $\tau \to \infty$, $k(\tau)$ tends to infinity and the BRD converges to the SFE which corresponds to the Walrasian equilibrium.

Analyze the BRD for the oligopoly where the generation capacity of each firm is Q and the marginal costs are zero. Prop. 4 below shows that in this case no monotone SBR exists under general assumptions on the initial bids. So we employ another approach. For a given linear demand $D(p) = \max\{0, \overline{D} - dp\}$, we study the best response dynamics depending on the ratio between $\overline{D}$ and Q. At every time $\tau = 1, 2, \ldots$, we search for the BR in the set of supply functions $S(p, \tau) = \min\{k(\tau)p, Q\}$. So the

problem is to find the optimal slope $k(\tau)$. Our aim is to determine ranges of parameter $\frac{\overline{D}}{Q}$ with the same best response dynamics.

Proposition 4 *The BRD for the model of n-firm oligopoly with $n \geq 3$ depends on the ratio between parameters $\overline{D}$ and Q as follows:*

For $\overline{D} > (n+1)Q$, $S(p,\tau) = \min\{Q, kdp\}$, $\forall k \in [1,\infty) \forall \tau \geq 1$. The BR is rather uncertain, but the outcome always corresponds to the Cournot equilibrium coinciding with the Walrasian equilibrium in this case.

For $(n-1)Q < \overline{D} < (n+1)Q$, the best response at the time $\tau = 1, 2, \ldots, T(\overline{D}, Q)$ is $S(p,\tau) = \min\{Q, k(\tau)p\}$, then the best response dynamics loops. The cycle length $T(\overline{D}, Q)$ is the smallest integer T meeting inequality $\sum_{s=0}^{T}(n-1)^s > \frac{4Q(\overline{D}-Q)}{(\overline{D}-(n-1)Q)^2}$;

For $\overline{D} < (n-1)Q$, $S(p,\tau) = \min\{Q, k(\tau)p\}$, $\forall \tau \geq 1$, where $k(\tau) = d\sum_{s=0}^{\tau-1}(n-1)^s$. For $\tau \to \infty$ the BRD converges to the Walrasian supply function.

Proof If $S(p,\tau) = \min\{Q, k(\tau)p\}$, then the residual demand at the time $\tau + 1$ is $D_{res}(p) = \max\{0, \overline{D} - dp - (n-1)\min\{Q, k(\tau)p\}\}$.

Denote $d(\tau) = d + (n-1)k(\tau)$ the maximal slope of the residual demand function. The form of the residual demand function depends on the ratio between Q and $\overline{D}$.

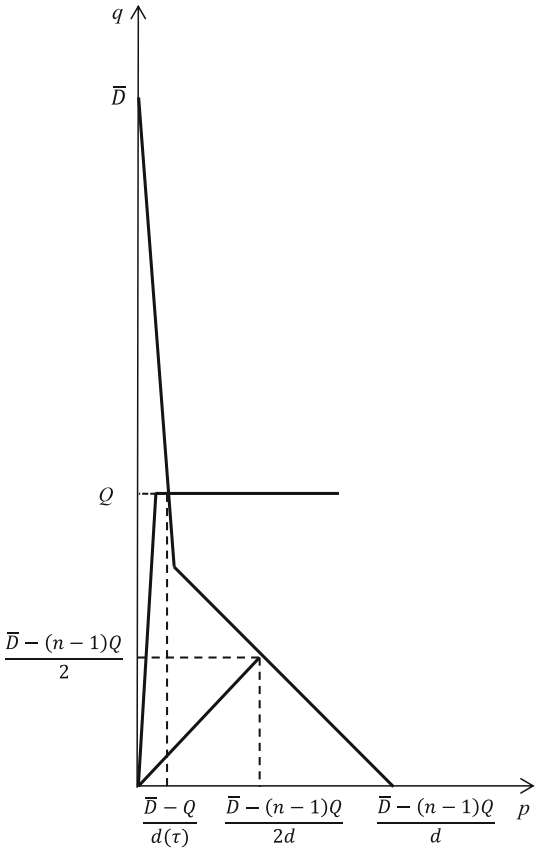


Fig. 3 The residual demand function and the two possible best responses for $\overline{D} \in (Q(n-1), Q(n+1))$

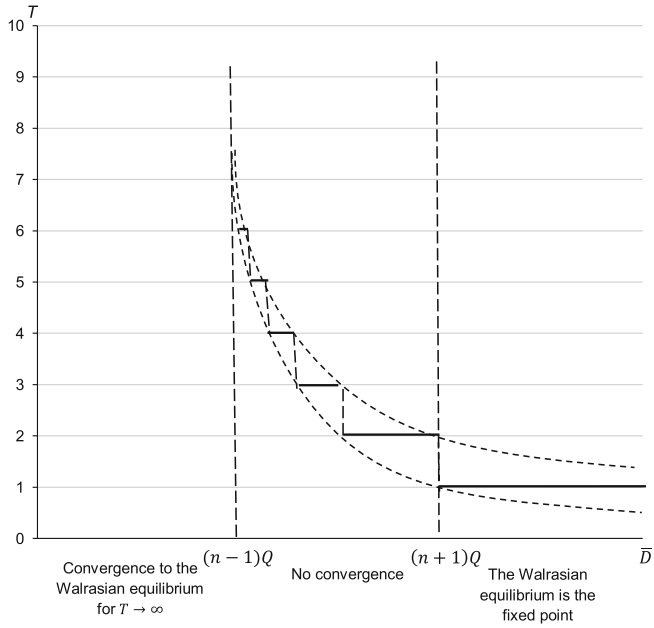


Fig. 4 The BRD cycle length T depending on the ratio between $\overline{D}$ and Q

There are two cases: (1) when the residual demand function has at time τ no kinks and (2) when it contains two linear segments (see Fig. 3).

The BRD corresponds to the first case if:

$$k(\tau)\overline{D} \leq d(\tau)Q. \tag{5}$$

For $\tau = 1$ the BR corresponds to the first case (as $k(0) = 0, d(0) = d$) under any $\overline{D}$ and is $\min\{Q, k(1)p\}$ for $k(1) = d$.

For $\overline{D} > (n+1)Q$, for any $k(\tau) = 0, d, 2d, \ldots$, the kink is above production capacity. The BR volume is Q, and the price $\tilde{p}$ setting the balance $\overline{D} - dp - (n-1)Q = Q$ is the Cournot equilibrium and the competitive equilibrium price.

For $\overline{D} < (n-1)Q$, there is a kink in the residual demand since inequality (5) holds for any $\tau = 0, 1, 2, \ldots$. At the time τ, $k(\tau) = d\sum_{s=0}^{\tau-1}(n-1)^s$, $d(\tau) = d\sum_{s=0}^{\tau}(n-1)^s$. The condition (5) becomes $\overline{D} \leq (n-1+\frac{d}{k(\tau)})Q$, and the BRD always meets the condition. For $\tau \to \infty$, the BRD converges to the SFE with the Walrasian outcome.

For $\overline{D} \in (Q(n-1), Q(n+1))$, there are two possible BR determined by the Cournot supply schedules (see Fig. 3). The first one is where $k(\tau+1) = d(\tau)$, $q^a(\tau) = Q$, $p(\tau) = \frac{\overline{D}-Q}{d(\tau)}$, $\pi_1(\tau) = \frac{(\overline{D}-Q)Q}{d(\tau)}$. The second one is where $k(\tau+1) = d$, $q^a(\tau) = \frac{\overline{D}-(n-1)Q}{2}$, $p(\tau) = \frac{\overline{D}-(n-1)Q}{2d}$, $\pi_2(\tau) = \frac{(\overline{D}-(n-1)Q)^2}{4d}$.

The first variant provides the greater profit $\pi_1(\tau) > \pi_2(\tau)$ if and only if $k(\tau+1) = d(\tau)$ is the optimal choice at the time τ, that is $\frac{d(\tau)}{d} \leq \frac{4Q(\overline{D}-Q)}{(\overline{D}-(n-1)Q)^2}$.

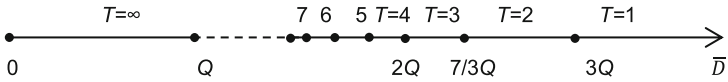


Fig. 5 The length of the BRD cycle depending on $\overline{D}$ value

For $\overline{D} \in (Q(n-1), Q(n+1))$, there always exists large enough τ such that $\pi_2(\tau) > \pi_1(\tau)$. Let $T(\overline{D}, Q)$ be the first time when this inequality holds. The BRD is cyclic with the length $T(\overline{D}, Q)$. For $t < T(\overline{D}, Q)$, $k(t) = d\sum_{s=0}^{\tau-1}(n-1)^s$ the BR at the time $T(\overline{D}, Q)$ is $k(T(\overline{D}, Q)+1) = d = d(1)$. The length of the cycle is the minimum integer number satisfying the inequality $\sum_{s=0}^{T}(n-1)^s > \frac{4q(\overline{D}-Q)}{(\overline{D}-(n-1)Q)^2}$. Thus, $T(\overline{D}, Q) = \lfloor \log_{(n-1)}\left(\frac{4Q(\overline{D}-Q)(n-2)}{(\overline{D}-(n-1)Q)^2}+1\right)\rfloor$, where $\lfloor x \rfloor$ is the largest integer not greater than x.

Figure 4 shows how the length of the BRD cycle varies depending on the ratio between $\overline{D}$ and the capacity constraint Q.

Figure 5 shows the intervals of $\overline{D}$ values with the same BRD for the duopoly.

Note that for $n = 2$ the difference is that, under $Q < \overline{D} < 2Q$, the first BR at time τ gives $q^a(\tau) = \frac{\overline{D}}{2}$, $\pi_1(\tau) = \frac{\overline{D}^2}{4d(\tau)}$.

5 Conclusions and discussion

Our study together with the previous research shows ambiguous results on the justification for the SFE concept. For market models with linear demand and marginal cost functions, the adaptive dynamics rapidly converges to the unique SFE existing under large demand shocks. A similar result holds if each generator either works all the time at its maximum capacity level, or never reaches this level, and its marginal cost is a linear function. Ayzenberg et al. (2013) show an interesting example of Siberian market that meets these assumptions. In this case the optimal auction design is to accept one bid for the whole period when the conditions hold, and to permit piece-wise linear bids or piece-wise constant bids with sufficiently many times. For the latter variant, Holmberg and Newbery (2010) show a possibility of SFE bid approximation.

However, in general the real costs of generating companies do not meet these assumptions. Typically such a company owns several generators with limited capacities (and in practice the constraints are often binding in spike periods). The price of fuel per 1MW is usually the main component of the marginal cost. Sometimes this cost is approximately constant, for other generators it is necessary to use a more complicated relation. But in any case the SFE bid is not an affine function since the capacity constraint is binding. While the literature focuses on the SFE specified in Theorem 1 as an instrument for the price forecast and evaluation of the market power in the supply function auction, our study shows that there is no ground to expect such kind of behavior in general.

For the symmetric oligopoly with zero marginal cost, limited total capacity nQ and maximum demand $\overline{D}$, Proposition 4 imposes that the SBRD exists and converges to the Walrasian equilibrium outcome if $\overline{D} < (n-1)Q$ or $\overline{D} > (n+1)Q$. However, for

each specified interval in the intermediate area $(n-1)Q < D < (n+1)Q$, the SBRD turns out to be cyclic with its special cycle. Under general assumptions on the demand shock distribution, the SBR does not exist and simple best response dynamics do not converge.

For similar markets, splitting of the time period, where a stable SFE does not exist, into short intervals with a certain demand and submission of a particular bid for each interval seems to be the better variant of the market architecture. Then a stable Nash equilibrium exists for each interval under general assumptions. Another possibility is to employ a different auction type, in particular, Vickrey auction. Vasin et al. (2007) show high efficiency of this auction by simulation of the market for the European zone of Russia.

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